

Balancing Predictive Performance and Interpretability in Clinical Risk Models:

A Comparative Study of Logistic Regression and Machine Learning Methods for Diabetes Prediction

Autumn Park

Abstract

Background: This study systematically compares the performance of logistic regression with contemporary machine learning algorithms for diabetes prediction. Compare and select the best model for predicting diabetes. while providing a comprehensive evaluation of the methodological complexities, inherent limitations, and empirical results associated with logistic regression and alternative approaches.

Objective: The purpose of the study is to evaluate whether logistic regression is comparable with more modern machine learning algorithms for diabetes prediction. This study emphasizes productive predictive performance calibration interpretation.

Methods: Using the Pima Indians Diabetes dataset, four models were developed and compared: logistic regression, LASSO-regularized logistic regression, random forest, and extreme gradient boosting (XGBoost). Data preprocessing included median imputation, feature standardization, and stratified train-test splitting. Model performance was evaluated using accuracy, sensitivity, specificity, F1 score, and area under the receiver operating characteristic curve (AUC). Calibration was assessed using reliability curves and Brier scores. Statistical comparisons of AUC values were conducted using DeLong's test. Model interpretability was evaluated using odds ratios, variable importance, and SHAP-style feature contributions.

Results: All models demonstrated comparable predictive performance, with AUC values ranging from 0.836 to 0.841. Logistic regression achieved an AUC of 0.837, random forest (0.840), and XGBoost (0.836). Differences in AUC were not statistically significant ($p > 0.05$). Calibration performance was similar across models, with Brier scores ranging from 0.154 to 0.158. Logistic regression has the most interpretable outputs, whereas machine learning models require additional explanatory methods.

Conclusion: Logistic regression remains a competitive and effective model for diabetes prediction in small clinical datasets, offering comparable predictive performance to machine

learning methods while maintaining superior interpretability. These findings highlight the importance of balancing accuracy and transparency in clinical decision support.

Keywords

Diabetes prediction; Logistic regression; Machine learning; Model interpretability; Calibration; Clinical decision support

Background

Diabetes mellitus (DM) is a global disease that impacts millions of individuals and contributes to cardiovascular disease and kidney failure. Early and accurate detection of diabetes is essential for effective prevention strategies, risk stratification, and the timely implementation of clinical interventions to mitigate disease progression and associated complications.

Recent advances in machine learning have led to increased adoption of complex predictive models, including gradient boosting, random forests, and neural networks. While these approaches may improve predictive accuracy, their limited interpretability presents challenges in clinical settings, where transparency and explainability are essential for decision-making.

Logistic regression remains a widely used method in clinical research due to its interpretability and ability to provide coefficients that can be translated into odds ratios. Prior work on predictive models for diabetes mellitus using machine learning techniques demonstrated that logistic regression achieved predictive performance comparable to that of gradient boosting models in a large clinical dataset, with AUC values of approximately 84.0% and 84.7%, respectively.

Building on this literature, the present study examines whether similar conclusions hold in smaller clinical datasets and extends prior work by incorporating calibration analysis, interpretability methods, and formal statistical comparisons.

Methods

Study Design

This study is a retrospective modeling analysis comparing multiple predictive algorithms for binary classification.

Data Source

The analysis used the Pima Indians Diabetes dataset, a widely used benchmark dataset for binary classification in healthcare. The dataset includes demographic and clinical variables, such as glucose level, body mass index (BMI), age, and insulin levels, as well as a binary outcome indicating diabetes status.

Data Preprocessing

Data preprocessing included:

- Median imputation for missing values
 - Conversion of outcome variable to binary factor
 - Stratified 80/20 train-test split
 - Feature standardization for models requiring scaled inputs
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Model Development

Four models were developed:

1. Logistic Regression
2. LASSO Logistic Regression (L1 regularization)
3. Random Forest (500 trees)
4. XGBoost (gradient boosting framework)

Hyperparameters were tuned using repeated 10-fold cross-validation.

Model Evaluation

Performance was evaluated using:

- Accuracy
- Sensitivity (recall)
- Specificity
- Precision
- F1 score
- Area Under the ROC Curve (AUC)

Calibration Analysis

Calibration was evaluated using:

- Reliability (calibration) curves
- Brier score

These metrics assess whether predicted probabilities correspond to observed outcome frequencies.

Statistical Analysis

Pairwise comparisons of AUC values were conducted using DeLong’s test to determine whether differences in model performance were statistically significant.

Interpretability Analysis

Model interpretability was evaluated using:

- Odds ratios (logistic regression)
 - Variable importance (random forest)
 - SHAP-style feature contributions (XGBoost)
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Results

Model Performance

All models demonstrated similar predictive performance:

Model	Accuracy	AUC
Logistic Regression	0.758	0.837
LASSO Logistic	0.758	0.841
Random Forest	0.752	0.840

XGBoost	0.745	0.836
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Differences in AUC were minimal (<0.01) and not statistically significant.

Calibration

Brier scores were comparable across models:

Model	Brier Score
Random Forest	0.154
LASSO Logistic	0.155
Logistic Regression	0.156
XGBoost	0.158

Calibration curves indicated that all models produced reasonably well-calibrated probability estimates.

Statistical Comparison

DeLong's test indicated no statistically significant differences between models ($p > 0.05$).

Feature Importance

Across all models, key predictors included:

- Glucose level
- Body mass index (BMI)
- Age
- Diabetes pedigree function

These findings are consistent with prior literature.

Interpretability

Logistic regression provided direct interpretability via odds ratios. In contrast, machine learning models require additional methods for explanation.

Discussion

This study shows that logistic regression stays competitive with modern machine learning methods in small clinical datasets. Despite the complexity of models such as random forest and XGBoost, these methods did not yield meaningful improvements in predictive performance.

These findings align with earlier research showing that simpler models can achieve performance comparable to that of more complex approaches in clinical prediction tasks.

A central implication is that model selection in healthcare should consider not only predictive accuracy but also explainability and calibration. Logistic regression offers a transparent and efficient approach that is well-suited for clinical decision support.

Limitations

This study has several limitations:

1. The dataset is relatively small and may limit generalizability
 2. The population is not representative of all demographic groups
 3. External validation was not performed
 4. Additional models (e.g., neural networks) were not evaluated
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Conclusion

Logistic regression remains a strong and competitive method for diabetes prediction. In this study, it achieved predictive performance comparable to modern machine learning models while offering superior interpretability.

These findings support the continued use of logistic regression as a baseline model in biomedical informatics, particularly in settings where transparency and clinical trust are essential.

Declarations (BMC Requirement)

Ethics approval and consent to participate: Not applicable (public dataset)

Consent for publication: Not applicable

Availability of data and materials: Publicly available dataset

Competing interests: None

Funding: None

Authors' contributions: AP conducted all analyses and manuscript preparation